

Gesture-Based Browser Shortcuts

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I PROBLEM STATEMENT AND MOTIVATION



- Traditional browsing relies on keyboard/ mouse clicks
- Accessibility barriers for users with limited motor skills
- Repetitive navigation (Back/Forward/Reload/New Tab) slows interaction.
- Goal → Make browsing faster, more intuitive, and gesture-controlled.

II PROPOSED SOLUTION

Gesture-Based Browser Shortcuts

- Users draw gestures anywhere on the page.
- Gestures are recognized via a \$1-like recognizer algorithm.
- Each gesture triggers browser actions:
 - Circle: Reload Tab
 - Left arrowhead: Back
 - Right arrowhead: Front
 - Z : New Tab
 - Up arrowhead: Home
 - Down arrowhead: End
- Works as overlay on any website (or integrated demo site)

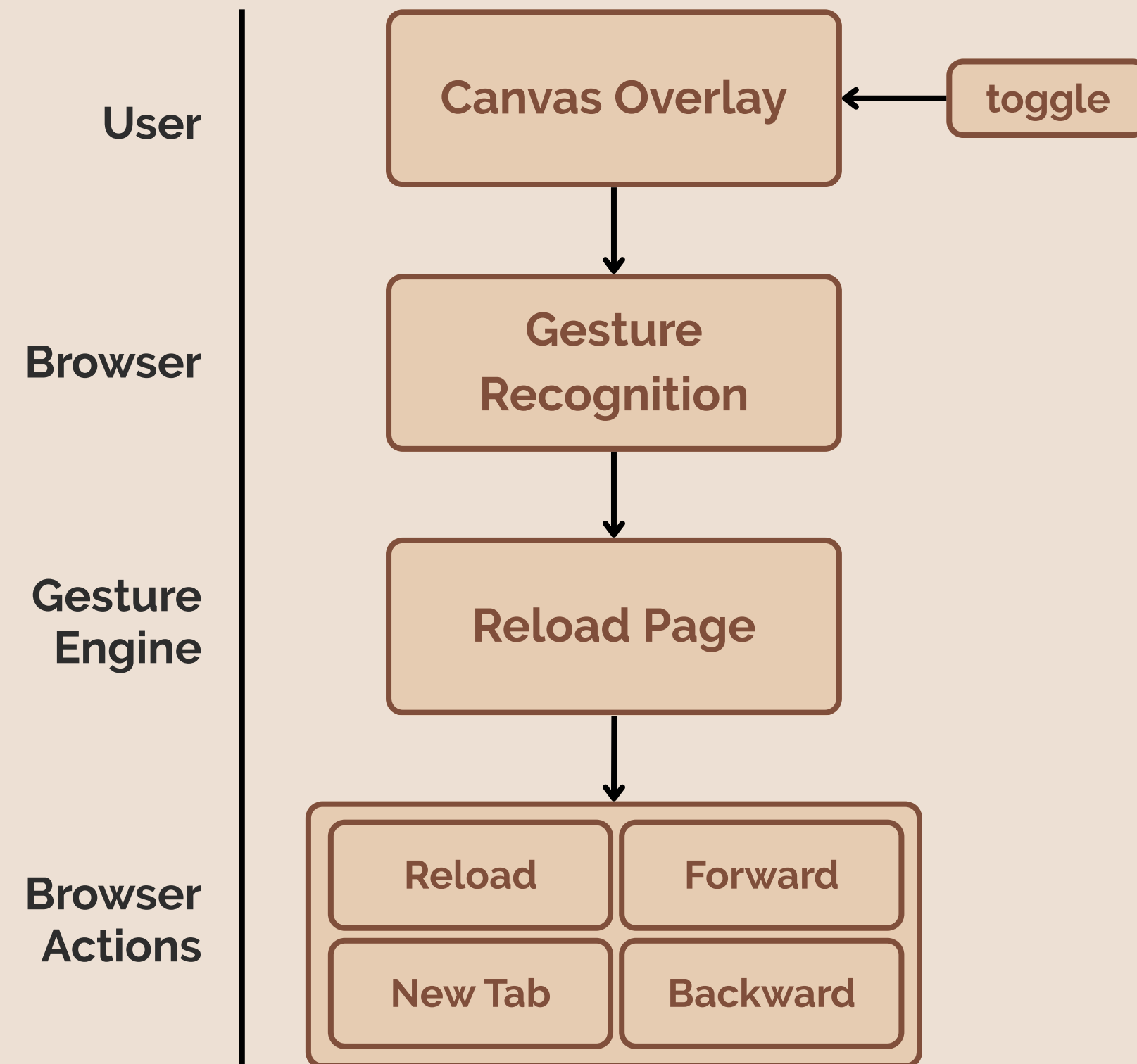


Small diagram of overlay canvas capturing gestures.

III IMPLEMENTATION OVERVIEW

Technology Stack

- HTML/CSS/JS front-end prototype
- Canvas API → captures stroke coordinates
- \$1 Recognizer → normalizes, resamples, compares patterns
- Non-blocking overlay → draw, scroll, and click simultaneously

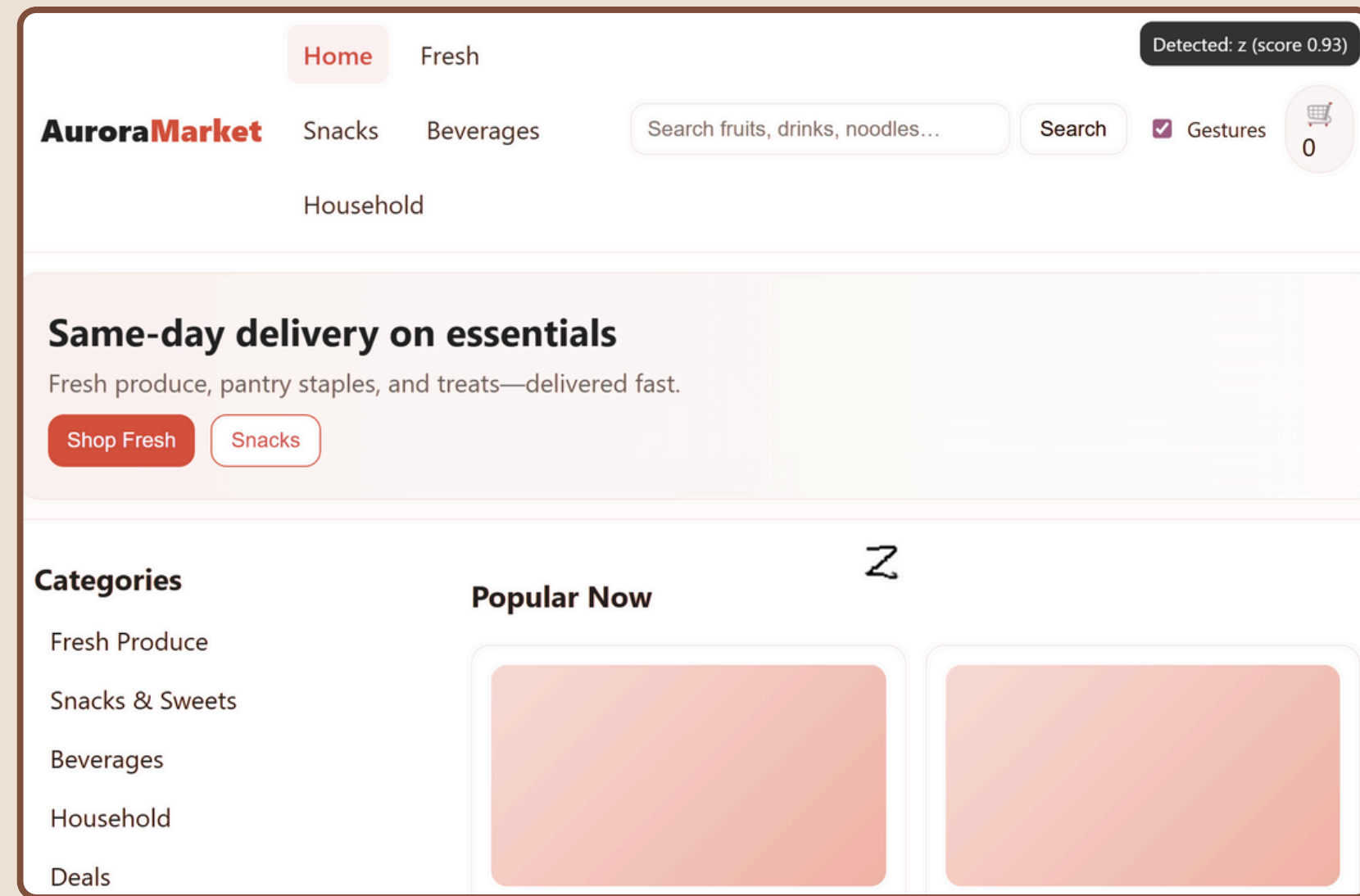


Architecture Block Diagram

IV DEMONSTRATION & KEY FEATURES

Technology Stack

- Full-screen drawing — gestures visible in high-contrast color
- No external libraries, pure JS
- Accurate recognition ($\approx 90\%$ +) and real-time feedback
- Gesture log & performance metrics (latency, accuracy)
- Lightweight $\rightarrow$ potential Chrome extension integration



Screenshot of the dummy site with a gesture trail overlay

IV RESULTS, LIMITATIONS & FUTURE WORK



Results

- Built a prototype with potential to help users with motor impairments like tremors or cerebral palsy.
- Offers alternative, low-precision way to surf websites.



Limitations

- Limited gesture vocabulary, may conflict with scrolls.
- User study limited — idea inspired by family members with motor impairments; broader testing pending.

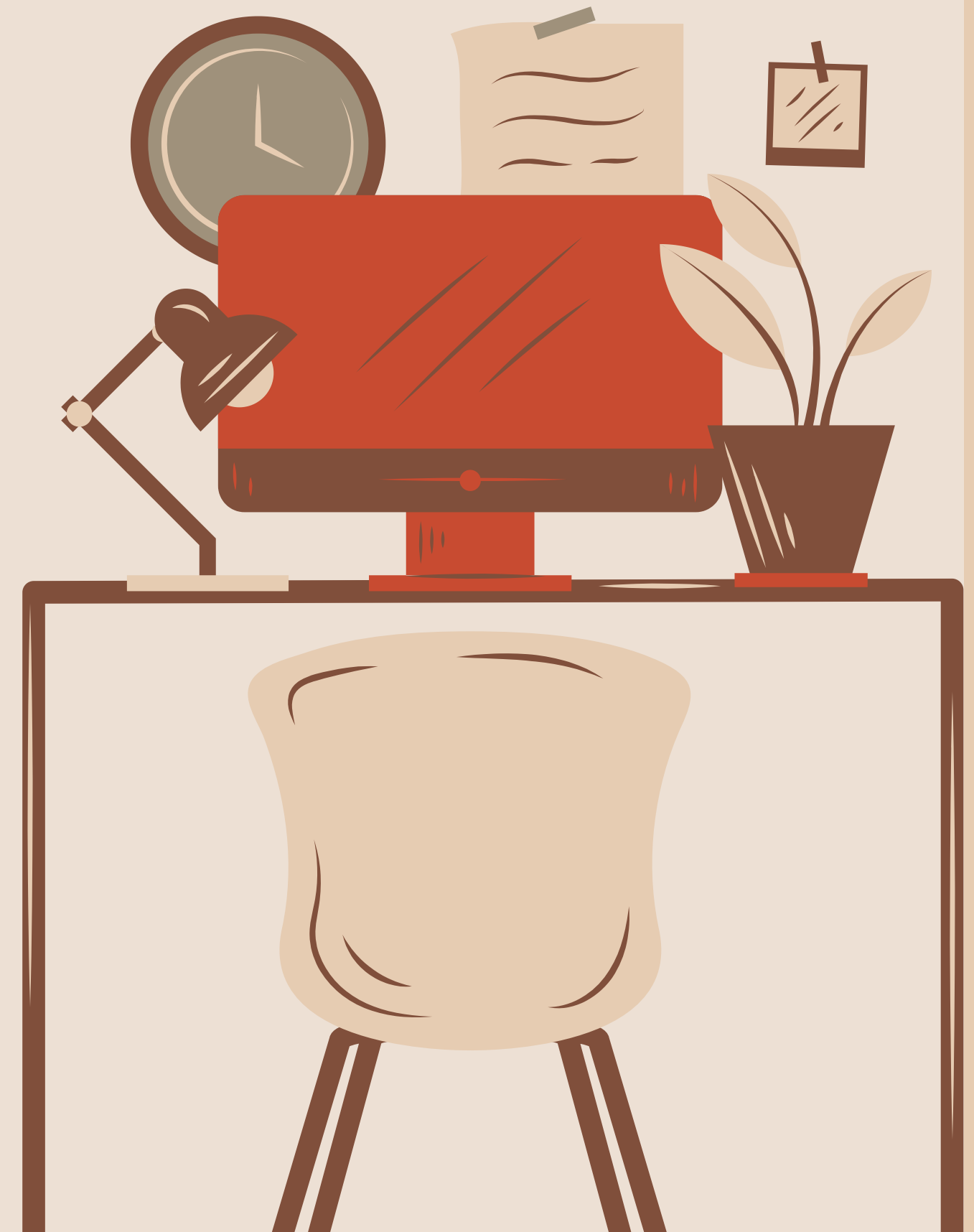


Future Work

- Publish system as open-source Chrome extension.
- Add more gesture commands
- Expand gesture customization for wider and different user needs.

V REFERENCES AND CREDITS

- Substantial help was taken from Perplexity AI for project development.
- The project idea was inspired by my grandmother with Parkinson's disease.
- YouTube tutorials were used to learn normalization, resampling, and pattern comparison using the \$1 Recognizer.
- superuser.com was very helpful in identifying suitable gesture mappings for each browser task.



Thank you for listening!

Suggestions ?

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